

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

CIRCLES

DIBY (DO IT BY YOURSELF)

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DIBY-01

The centre of circle inscribed in square formed by the lines $x^2 - 8x + 12 = 0$ and $y^2 - 14y + 45 = 0$, is

[IIT-JEE-2003]

- (A) (4, 7) (B) (7, 4) (C) (9, 4) (D) (4, 9)

[Ans.: (A)]

DIBY-02

Let AB be a chord of the circle $x^2 + y^2 = r^2$ subtending a right angle at the center. Then, the locus of the centroid of the $\triangle PAB$ as P moves on the circle, is

[IIT-JEE 2001]

- (A) a parabola (B) a circle
(C) an ellipse (D) a pair of straight lines

[Ans.: (B)]

DIBY-03

The lines $2x - 3y = 5$ and $3x - 4y = 7$ are diameters of a circle of area 154 sq units. Then, the equation of this circle is

[IIT-JEE 1989]

- (A) $x^2 + y^2 + 2x - 2y = 62$ (B) $x^2 + y^2 + 2x - 2y = 47$
(C) $x^2 + y^2 - 2x + 2y = 47$ (D) $x^2 + y^2 - 2x + 2y = 62$

[Ans.: (C)]

DIBY-04

Four distinct points $(2k, 3k)$, $(1, 0)$, $(0, 1)$ and $(0, 0)$ lie on a circle for k equal to:

[JEE Mains-2024]

(A) $\frac{2}{13}$

(B) $\frac{3}{13}$

(C) $\frac{5}{13}$

(D) $\frac{1}{13}$

[Ans.: (C)]

DIBY-05

Equation of two diameters of a circle are $2x - 3y = 5$ and $3x - 4y = 7$. The line joining the points $\left(-\frac{22}{7}, -4\right)$ and $\left(-\frac{1}{7}, 3\right)$ intersects the circle at only one point $P(\alpha, \beta)$. Then $17\beta - \alpha$ is equal to

[JEE Mains-2024]

[Ans.:]

DIBY-06

The points of intersection of the line $ax + by = 0$, ($a \neq b$) and the circle $x^2 + y^2 - 2x = 0$ are $A(\alpha, 0)$ and $B(1, \beta)$. The image of the circle with AB as a diameter in the line $x + y + 2 = 0$ is :

[JEE Mains-2023]

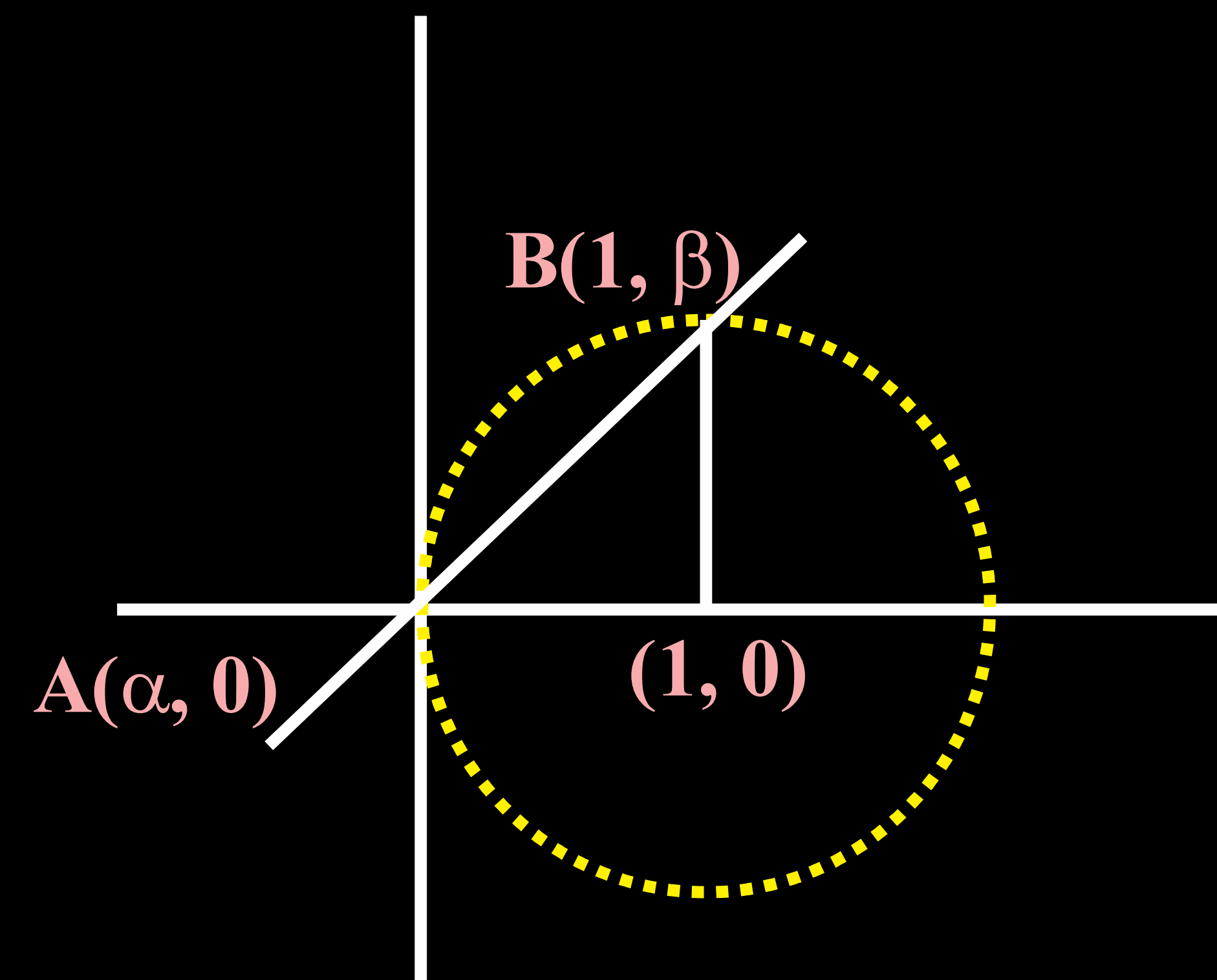
[Ans.: A]

A $x^2 + y^2 + 5x + 5y + 12 = 0$

B $x^2 + y^2 + 3x + 5y + 8 = 0$

C $x^2 + y^2 + 3x + 3y + 4 = 0$

D $x^2 + y^2 - 5x - 5y + 12 = 0$



DIBY-07

The equation of the circle passing through the points $(1, 0)$ and $(0, 1)$ and having the smallest radius is **[AIEEE-2011]**

(A) $x^2 + y^2 + x + y - 2 = 0$

(B) $x^2 + y^2 - 2x - 2y + 1 = 0$

(C) $x^2 + y^2 - x = 0$

(D) $x^2 + y^2 + 2x + 2y - 7 = 0$

[Ans. : (C)]

DIBY-08

The circle passing through $(1, -2)$ and touching the axis of x at $(3, 0)$ also passes through the point **[JEE Mains-2013]**

(A) $(-5, 2)$

(B) $(2, -5)$

(C) $(5, -2)$

(D) $(-2, 5)$

[Ans. : (C)]

DIBY-09

Equation of the tangent the circle $x^2 + y^2 - 3x + 4y - 5 = 0$ at the point $(3, 1)$ is?

- (A) $2x + y - 7 = 0$ (B) $x + 2y - 5 = 0$
(C) $3x + 2y - 11 = 0$ (D) $3x + 4y - 13 = 0$

[Ans.: (B)]

DIBY-10

Let C be a circle with radius $\sqrt{10}$ units and centre at the origin. Let the line $x + y = 2$ intersects the circle C at the points P and Q . Let MN be a chord of C of length 2 unit and slope -1 . Then, a distance (in units) between the chord PQ and the chord MN is:

[JEE Mains-2024]

- (A) $3 - \sqrt{2}$ (B) $2 - \sqrt{3}$ (C) $\sqrt{2} - 1$ (D) $\sqrt{2} + 1$

[Ans.: (A)]

DIBY-11

The tangent and the normal lines at the point $(\sqrt{3}, 1)$ to the circle $x^2 + y^2 = 4$ and the x -axis form a triangle. The area of this triangle (in square units) is

[JEE Mains-2019]

- (A) $\frac{1}{3}$ (B) $\frac{4}{\sqrt{3}}$ (C) $\frac{2}{\sqrt{3}}$ (D) $\frac{1}{\sqrt{3}}$

[Ans.: (C)]



DIBY-12

Let a circle C of radius 1 and closer to the origin be such that the lines passing through the point (3, 2) and parallel to the coordinate axes touch it. Then the shortest distance of the circle C from the point (5, 5) is: **[JEE Mains-2024]**

- (A) $4\sqrt{2}$ (B) 4 (C) 5 (D) $2\sqrt{2}$

[Ans. : (B)]

DIBY-13

Consider a circle $(x - \alpha)^2 + (y - \beta)^2 = 50$ where $\alpha, \beta > 0$. If the circle touches the line $y + x = 0$ at the point P, whose distance from the origin is $4\sqrt{2}$, then $(\alpha + \beta)^2$ is equal to

[JEE Mains-2024]

[Ans. : 100]

DIBY-14

A line segment AB of length λ moves such that the points A and B remain on the periphery of a circle of radius λ . Then the locus of the point, that divides the line segment AB in the ratio 2 : 3, is a circle of radius. **[JEE Mains-2023]**

- (A) $\frac{2}{3}\lambda$ (B) $\frac{3}{5}\lambda$ (C) $\frac{\sqrt{19}}{7}\lambda$ (D) $\frac{\sqrt{19}}{5}\lambda$

[Ans. : (D)]

DIBY-15

The equation of the locus of the mid-points of chords of the circle $4x^2 + 4y^2 - 12x + 4y + 1 = 0$, which subtend angle 120° at the centre, is the circle

- (A) $x^2 + y^2 - 3x + y + \frac{31}{16} = 0$ (B) $x^2 + y^2 - 3x + y - \frac{1}{16} = 0$
 (C) $x^2 + y^2 - 3x + y + \frac{1}{4} = 0$ (D) $x^2 + y^2 - 3x + y - \frac{1}{4} = 0$

[Ans. : (C)]

DIBY-16

Let the locus of the mid points of the chords of circle $x^2 + (y-1)^2 = 1$ drawn from the origin intersect the line $x + y = 1$ at P and Q. Then, the length of PQ is **[JEE Mains-2024]**

- (A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) $1/2$ (D) 1

[Ans. : (A)]



DIBY-17

A circle is inscribed in an equilateral triangle of side of length 12. If the area and perimeter of any square inscribed in this circle are m and n , respectively, then $m + n^2$ is equal to **[JEE Mains-2024]**

- (A) 408 (B) 414 (C) 312 (D) 396

[Ans. : (A)]

DIBY-18

The sum of the squares of the lengths of the chords intercepted on the circle, $x^2 + y^2 = 16$, by the lines, $x + y = n$, $n \in N$, where N is the set of all natural numbers, is

[JEE Mains-2019]

- (A) 210 (B) 160 (C) 105 (D) 320

[Ans. : (A)]

DIBY-19

The tangent to the circle $C_1 : x^2 + y^2 - 2x - 1 = 0$ at the point $(2, 1)$ cuts off a chord of length 4 from a circle C_2 where centre is $(3, -2)$. The radius of C_2 is

[JEE Mains-2018]

- (A) 2 (B) $\sqrt{2}$ (C) 3 (D) $\sqrt{6}$

[Ans. :]

DIBY-20

If a variable line, $3x + 4y - \lambda = 0$ is such that the two circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 18x - 2y + 78 = 0$ are on its opposite sides, then the set of all values of λ is the interval.

[JEE Mains-2019]

- (A) (23, 31) (B) (13, 23) (C) (2, 17) (D) (12, 21)

[Ans. (D)]

DIBY-21

Point of contact of line $y = x + 4\sqrt{2}$ to the circle $x^2 + y^2 = 16$ is

- (A) $(2\sqrt{2}, -2\sqrt{2})$ (B) $(-2\sqrt{2}, -2\sqrt{2})$
 (C) $(-2\sqrt{2}, 2\sqrt{2})$ (D) $(2\sqrt{2}, 2\sqrt{2})$

[Ans. : (C)]

DIBY-22

Equation of circle whose centre is $(3, -1)$ and which cut off an intercept of length 6 unit from the line $2x - 5y + 18 = 0$ is

- (A) $x^2 + y^2 - 6x + 2y - 28 = 0$ (B) $x^2 + y^2 - 6x + 2y - 11 = 0$
 (C) $x^2 + y^2 - 6x + 2y - 18 = 0$ (D) $x^2 + y^2 - 6x + 2y - 27 = 0$

[Ans. : (A)]

DIBY-23

The circle $x^2 + y^2 - 6x - 10y + k = 0$ does not touch or intersect x -axis but cuts y -axis at two distinct points. Find the ranges of the values of k .

[Ans. : $9 < k < 25$]



DIBY-24

A ray of light incident at the point $(-2, -1)$ gets reflected from the tangent at $(0, -1)$ to the circle $x^2 + y^2 = 1$. If the reflected ray touches the circle, then find the equation of the line along which the incident ray moved.

$$[\text{Ans. : } 4x + 3y + 11 = 0]$$

DIBY-25

If a chord of the circle $x^2 + y^2 - 4x - 2y - c = 0$ is trisected at the points $(\frac{1}{3}, \frac{1}{3})$ and $(\frac{8}{3}, \frac{8}{3})$, then find c .

$$[\text{Ans. : } 20]$$

DIBY-26

Angle between the tangents drawn from a point P to the circle $x^2 + y^2 - 4x - 8y + 8 = 0$ is 60° , then find the length of the tangent drawn from P to the circle.

$$[\text{Ans. : } 06]$$

DIBY-27

Find the co-ordinates of the point of intersection of tangents at the points, where the line $2x + y + 12 = 0$ meets the circle $x^2 + y^2 - 4x + 3y - 1 = 0$.

$$[\text{Ans. : } (1, -2)]$$



DIBY-28

From a point P outside a circle with centre at C , tangents PA and PB are drawn such that $\frac{1}{(CA)^2} + \frac{1}{(PA)^2} = \frac{1}{16}$, then find the length of chord AB .

[Ans. : 08]

DIBY-29

The number of common tangents to the circles $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 18y + 26 = 0$, is

[JEE (Main)-2015]

- (A) 1 (B) 2 (C) 3 (D) 4

[Ans. : (C)]

DIBY-30

Let the orthocentre and centroid of a triangle be $A(-3, 5)$ and $B(3, 3)$ respectively. If C is the circumcentre of this triangle, then the radius of the circle having line segment AC as diameter, is

[JEE (Main)-2018]

- (A) $\sqrt{10}$ (B) $2\sqrt{10}$ (C) $3\sqrt{\frac{5}{2}}$ (D) $\frac{3\sqrt{5}}{2}$

[Ans. : (C)]

DIBY-31

Paragraph for Q.Nos. A and B

A tangent PT is drawn to the circle $x^2 + y^2 = 4$ at the point $P(\sqrt{3}, 1)$. A straight line L , perpendicular to PT is a tangent to the circle $(x - 3)^2 + y^2 = 1$.

Choose the correct answer:

[IIT-JEE-2012 (Paper-2)]

A. A common tangent of the two circles is

(A) $x = 4$

(B) $y = 2$

(C) $x + \sqrt{3}y = 4$

(D) $x + 2\sqrt{2}y = 6$

[Ans. : (D)]

B. A possible equation of L is

(A) $x - \sqrt{3}y = 1$

(B) $x + \sqrt{3}y = 1$

(C) $x - \sqrt{3}y = -1$

(D) $x + \sqrt{3}y = 5$

[Ans. : (A)]

DIBY-32

Let the locus of the mid points of the chords of circle $x^2 + (y - 1)^2 = 1$ drawn from the origin intersect the line $x + y = 1$ at P and Q . Then, the length of PQ is **[JEE Mains-2024]**

- (A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) $\frac{1}{2}$ (D) 1

[Ans. : (A)]

DIBY-33

If the tangents at the points P and Q on the circle $x^2 + y^2 - 2x + y = 5$ meet at the point $R\left(\frac{9}{4}, 2\right)$, then the area of the triangle PQR is **[JEE Mains-2023]**

- (A) $\frac{13}{4}$ (B) $\frac{13}{8}$ (C) $\frac{5}{4}$ (D) $\frac{5}{8}$

[Ans. : (D)]

DIBY-34

Let the tangents drawn from the origin to the circle, $x^2 + y^2 - 8x - 4y + 16 = 0$ touch it at the points A and B . Then $(AB)^2$ is equal to **[JEE Mains-2020]**

- (A) $\frac{56}{5}$ (B) $\frac{32}{5}$ (C) $\frac{64}{5}$ (D) $\frac{52}{5}$

[Ans. : (C)]

DIBY-35

The number of common tangents, to the circle $x^2 + y^2 - 18x - 15y + 131 = 0$ and $x^2 + y^2 - 6x - 6y - 7 = 0$, is

[JEE Mains-2023]

- (A) 3 (B) 2 (C) 1 (D) 4

[Ans. : (A)]

DIBY-36

Let $C : x^2 + y^2 = 4$ and $C' : x^2 + y^2 - 4\lambda x + 9 = 0$ be two circles. If the set of all values of λ so that the circles C and C' intersect at two distinct points, is $R - [a, b]$, then the point $(8a + 12, 16 - 20)$ lies on the curve.

[JEE Mains-2024]

- (A) $x^2 + 2y^2 - 5x + 6y = 3$ (B) $5x^2 - y = -11$
(C) $x^2 - 4y^2 = 7$ (D) $6x^2 + y^2 = 42$

[Ans. : (D)]

DIBY-37

If the circles $x^2 + y^2 + 6x + 8y + 16 = 0$ and $x^2 + y^2 + 2(2 - \sqrt{3})x + 2(4 - \sqrt{6})y = k + 6\sqrt{3} + 8\sqrt{6}$, $k > 0$ touch internally at the point $P(\alpha, \beta)$, then $(\alpha + \sqrt{3})^2 + (\beta + \sqrt{6})^2$ is equal to

[JEE Mains-2022]

[Ans. : 25]

DIBY-38

Let r_1 and r_2 be the radii of the largest and smallest circles, respectively, which pass through the point $(-4, 1)$ and having their centres on the circumference of the circle $x^2 + y^2 + 2x + 4y - 4 = 0$. If $\frac{r_1}{r_2} = a + b\sqrt{2}$, then $a + b$ is equal to [JEE Mains-2021]

- (A) 5 (B) 11 (C) 7 (D) 3

[Ans. : (A)]

DIBY-39

Let the line $x + y = 1$ meet the circle $x^2 + y^2 = 4$ at the points A and B . If the line perpendicular to AB and passing through the mid point of the chord AB intersects the circle at C and D , then the area of the quadrilateral $ADBC$ is equal to [JEE Mains-2025]

- (A) $3\sqrt{7}$ (B) $\sqrt{14}$ (C) $2\sqrt{14}$ (D) $5\sqrt{7}$

[Ans. : (c)]

DIBY-40

The distance between the chords of contact of the tangents to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ from the origin and from the point (g, f) is

- (A) $g^2 + f^2$ (B) $\frac{1}{2}(g^2 + f^2 + c)$ (C) $\frac{1}{2} \frac{g^2 + f^2 + c}{\sqrt{g^2 + f^2}}$ (D) $\frac{1}{2} \frac{(g^2 + f^2 - c)}{\sqrt{g^2 + f^2}}$

[Ans. : (D)]

DIBY-41

A circle with centre $(2, 3)$ and radius 4 intersects the line $x + y = 3$ at the points P and Q . If the tangents at P and Q intersect at the points $S(\alpha, \beta)$, then $4\alpha - 7\beta$ is [JEE Mains-2023]

[Ans. : 11]

DIBY-42

The locus of the middle points of those chords of the circle $x^2 + y^2 = 4$ which subtend a right angle at the origin is

- (A) $x^2 + y^2 - 2x - 2y = 0$ (B) $x^2 + y^2 = 4$
(C) $x^2 + y^2 = 2$ (D) $(x - 1)^2 + (y - 2)^2 = 5$

[Ans. : (C)]

DIBY-43

The area of the triangle formed by the tangents from the points (h, k) to the circle $x^2 + y^2 = a^2$ and the line joining their points of contact is

(A) $a \frac{(h^2 + k^2 - a^2)^{3/2}}{h^2 + k^2}$

(B) $a \frac{(h^2 + k^2 - a^2)^{1/2}}{h^2 + k^2}$

(C) $\frac{(h^2 + k^2 - a^2)^{3/2}}{h^2 + k^2}$

(D) $\frac{(h^2 + k^2 - a^2)^{1/2}}{h^2 + k^2}$

[Ans. : (A)]

DIBY-44

Let the circles $C_1: (x - \alpha)^2 + (y - \beta)^2 = r_1^2$ and $C_2: (x - 8)^2 + \left(y - \frac{15}{2}\right)^2 = r_2^2$ touch each other externally at the point $(6, 6)$. If the point $(6, 6)$ divides the line segment joining the centres of the circles C_1 and C_2 internally in the ratio $2 : 1$, then $(\alpha + \beta) + 4(r_1^2 + r_2^2)$ equals

[JEE Mains-2024]

(A) 110

(B) 130

(C) 125

(D) 145

[Ans. : (B)]

DIBY-45

Choose the incorrect statement about the two circles whose equations are given below:

$$x^2 + y^2 - 10x - 10y + 41 = 0 \text{ and } x^2 + y^2 - 16x - 10y + 80 = 0.$$

[JEE Mains-2021]

- (A) Distance between two centres is the average of radii of both the circles.
- (B) Both circles' centres lie inside region of one another.
- (C) Circles have two intersection points.
- (D) Both circles pass through the centre of each other.

[Ans. :]

DIBY-46

If one of the diameters of the circle $x^2 + y^2 - 2\sqrt{2}x - 6\sqrt{2}y + 14 = 0$ is a chord of circle

$$(x - 2\sqrt{2})^2 + (y - 2\sqrt{2})^2 = r^2, \text{ then the value of } r^2 \text{ is equal to}$$

[JEE Mains-2022]

[Ans. : 10]

DIBY-47

The set of values of k , for which the circle $C : 4x^2 + 4y^2 - 12x + 8y + k = 0$ lies inside the fourth quadrant and the point $\left(1, -\frac{1}{3}\right)$ lie on or inside the circle C , is

[JEE Mains-2022]

- (A) An empty set
- (B) $\left(6, \frac{65}{9}\right]$
- (C) $\left[\frac{80}{9}, 10\right)$
- (D) $\left(9, \frac{92}{9}\right]$

[Ans. : (D)]

DIBY-48

If the circles $x^2 + y^2 + 6x + 8y + 16 = 0$ and $x^2 + y^2 + 2(3 - \sqrt{3})x + 2(4 - \sqrt{6})y = k + 6\sqrt{3} + 8\sqrt{6}$, $k > 0$, touch internally at the point $P(\alpha, \beta)$, then $(\alpha + \sqrt{3})^2 + (\beta + \sqrt{6})^2$ is equal to

[JEE Mains-2022]

[Ans. : 25]

DIBY-49

Let $C : x^2 + y^2 = 4$ and $C' : x^2 + y^2 - 4\lambda x + 9 = 0$ be two circles. If the set of all values of λ so that the circles C and C' intersect at two distinct points, is $R - [a, b]$, then the point $(8a + 12, 16 - 20)$ lies on the curve.

[JEE Mains-2024]

(A) $x^2 + 2y^2 - 5x + 6y = 3$

(B) $5x^2 - y = -11$

(C) $x^2 - 4y^2 = 7$

(D) $6x^2 + y^2 = 42$

[Ans. : (D)]

DIBY-50

The equation of the circle through the points of intersection of $x^2 + y^2 - 1 = 0$, $x^2 + y^2 - 2x - 4y + 1 = 0$ and touching the line $x + 2y = 0$, is

(A) $x^2 + y^2 + x + 2y = 0$

(B) $x^2 + y^2 - x + 20 = 0$

(C) $x^2 + y^2 - x - 2y = 0$

(D) $2(x^2 + y^2) - x - 2y = 0$

[Ans. : (C)]

DIBY-51

Two circles each of radius 5 units touch each other at the point $(1, 2)$. If the equation of their common tangent is $3x + 4y = 10$, and $C_1(\alpha, \beta)$ and $C_2(\gamma, \delta)$, $C_1 \neq C_2$ are their centres, then $|[\alpha + \beta](\gamma + \delta)|$ equal to

[JEE Mains-2021]

[Ans. : 40]

DIBY-52

If a circle passes through the point $(1, 2)$ and cuts the circle $x^2 + y^2 = 4$ orthogonally, then the equation of the locus of its centre is

(A) $x^2 + y^2 - 3x - 8y + 1 = 0$

(B) $x^2 + y^2 - 2x - 6y - 7 = 0$

(C) $2x + 2y - 9 = 0$

(D) $2x + 4y - 1 = 0$

[Ans. : (C)]

DIBY-53

Let the centre of a circle, passing through the points $(0, 0)$, $(1, 0)$ and touching the circle $x^2 + y^2 = 9$, be (h, k) . Then for all possible values of the coordinates of the centre (h, k) , $4(h^2 + k^2)$ is equal to

[JEE Mains-2024]

[Ans. : 9]

DIBY-54

The radical centre of the circles $x^2 + y^2 + 4x + 6y = 19$, $x^2 + y^2 = 9$ and $x^2 + y^2 - 2x - 2y = 5$ will be [Ans.: A]

- (A) (1, 1) (B) (-1, 1) (C) (1, -1) (D) (0, 1)

DIBY-55

Find the radical centre of circles $x^2 + y^2 + 3x + 2y + 1 = 0$, $x^2 + y^2 - x + 6y + 5 = 0$ and $x^2 + y^2 + 5x - 8y + 15 = 0$. Also find the equation of the circle cutting them orthogonally. [Ans.: (3, 2), $(x - 3)^2 + (y - 2)^2 = 27$]

DIBY-56

Radical centre of the circles drawn on the sides as diameter of triangle formed by lines $3x - 4y + 6 = 0$, $x - y + 2 = 0$ and $4x + 3y - 17 = 0$. [Ans.: D]

- (A) (3, 2) (B) (3, -2) (C) (2, -3) (D) (2, 3)

DIBY-57

A circle C touches the line $x = 2y$ at the point (2, 1) and intersects the circle $C_1: x^2 + y^2 + 2y - 5 = 0$ at two points P and Q such that PQ is a diameter of C_1 . Then the diameter of C is

[JEE Mains-2021]

[Ans. : A]

- (A) $7\sqrt{5}$ (B) 15 (C) $\sqrt{285}$ (D) $6\sqrt{15}$

DIBY-58

Triangle ABC is right angled at A. The circle with centre A and radius AB cuts BC and AC internally at D and E respectively. If $BD = 20$ and $DC = 16$ then find the length AC.

[Ans. : $6\sqrt{26}$]

DIBY-59

Consider a circle C which touches the y-axis at (0, 6) and cuts off an intercept $6\sqrt{5}$ on the x-axis. Then the radius of the circle C is equal to:

[JEE Mains-2021]

[Ans. : B]

- (A) $\sqrt{53}$ (B) 9 (C) 8 (D) $\sqrt{82}$

DIBY-60

Two tangents are drawn from the point $P(-1, 1)$ to the circle $x^2 + y^2 - 2x - 6y + 6 = 0$. If these tangents touch the circle at points A and B, and if D is a point on the circle such that length of the segments AB and AD are equal, then the area of the triangle ABD is equal to:

[JEE Mains-2021]

[Ans. : c]

- (A) 2 (B) $(3\sqrt{2} + 2)$ (C) 4 (D) $3(\sqrt{2} - 1)$

DIBY-61

Let r_1 and r_2 be the radii of the largest and smallest circles, respectively, which pass through the point $(-4, 1)$ and having their centres on the circumference of the circle $x^2 + y^2 + 2x + 4y - 4 = 0$. If $\frac{r_1}{r_2} = a + b\sqrt{2}$, then $a + b$ is equal to:

[JEE Mains-2021]

- (A) 3 (B) 1 (C) 5 (D) 7 [Ans. : c]

DIBY-62

Show that the equation of a straight line meeting the circle $\sqrt{x^2 + y^2} = a^2$ in two points at equal distance d from a point (x_1, y_1) on its circumference is $xx_1 + yy_1 - a^2 + (d^2/2) = 0$.



*It is never about **end result**
It's always about the **Journey***

THANKYOU

#future **IITians**